Mobile Money User Activity Classification Report

Executive Summary

This project constructs an end-to-end pipeline to classify mobile money users into low, medium, and high

Key deliverables:
- Cleaned transaction dataset from raw SMS exports.
- User-level feature engineering and segmentation.
- EDA visualizations that highlight usage patterns.
- Comparative modeling and selection of the best classifier.
- Interpretation and business recommendations for feature-driven engagement.

Introduction

Mobile money services are increasingly important in financial inclusion and daily transaction behavior. Th

Data Collection Methodology

Raw SMS exports were collected from MobileMoney and OrangeMoney sources, then paired with projec

Privacy measures:
- All user identifiers are anonymized.
- No personal identifiers are included in the cleaned dataset.
- Data storage and handling follow secure project standards.

Data Cleaning and Preparation

The cleaning process transformed raw SMS logs into an analysis-ready dataset by:
- Normalizing headers from both `.csv` and `.xlsx` sources.
- Standardizing dates and text content.
- Tagging transaction types and failed status from message text.
- Removing OTP, promotional messages, and duplicate records.
- Preserving activity signals from balance checks, adjustments, and failed attempts.

The cleaned dataset contains 2,261 rows out of 25,996 raw records, representing a focused set of valid r

Exploratory Data Analysis

EDA reveals the following patterns:
- User activity is imbalanced, with a minority of users driving the highest volume.
- Active users show a strong relationship between transaction frequency and total volume.
- Failed transactions occur more often among higher activity users, suggesting retry behavior or friction.
- Weekend activity is a distinguishing pattern for medium and high activity segments.
- Correlation analysis confirms that normalized activity features are effective for segmentation.

Modeling Approach

Three classifiers were evaluated:
- Logistic Regression
- Decision Tree
- Random Forest

A macro F1 score was chosen as the primary evaluation metric because it treats each activity class equa

Feature preparation included:
- Scaling numeric features with `StandardScaler`.

- One-hot encoding categorical features where needed.
- Dropping metadata columns such as `user_id` and threshold values.

Results and Interpretation

Model performance results:
- Decision Tree: accuracy = 0.6667, precision (macro) = 0.50, recall (macro) = 0.6667, F1 (macro) = 0.55
- Random Forest: accuracy = 0.6667, precision (macro) = 0.50, recall (macro) = 0.6667, F1 (macro) = 0.5
- Logistic Regression: accuracy = 0.0, precision = 0.0, recall = 0.0, F1 = 0.0

Based on these results, the Decision Tree model was selected for its strong balanced performance and

Interpretation and Key Drivers

Top feature drivers include:
- `total_transaction_volume`
- `activity_score`

These features summarize sustained usage and user commitment, making them reliable predictors of se

Business Recommendations

The analysis supports the following actions:
- Offer retention incentives to high-volume, high-frequency users.
- Design merchant or payment bundles for customers with frequent payment behavior.
- Monitor failed transaction attempts to reduce friction and improve the customer experience.
- Use weekend activity trends to tailor promotions for after-hours mobile money usage.
- Provide educational support for low activity users to increase adoption.

Limitations and Ethics

Limitations:
- The sample size is limited and model performance may vary with more data.
- SMS parsing can misclassify edge cases due to mixed language and varying message formats.
- The model is based solely on transactional behavior and does not include demographic or contextual fe

Ethical guidance:
- Use model outputs as segmentation signals, not as final decisions.
- Avoid deploying predictions for credit scoring or other high-stakes uses without further validation.
- Maintain data confidentiality and avoid re-identification risk.

Conclusion

This submission delivers a complete analytic workflow from raw SMS exports to user-level classification a